

Question 1 (14 marks) **Start a new booklet**

(a) Simplify $4\sqrt{45} - 3\sqrt{54} - 6\sqrt{5}$ ✓ (2)

(b) Solve for y : $6(y-1) = 3(y+8)$ ✓ (2)

(c) Solve simultaneously: $5x - 2y = -1$ (3)
 $8x - y = 5$

(d) Evaluate $\sqrt[3]{\frac{84 \times 69}{35 + 37}}$. Round your answer to three significant figures. (2)

(e) Evaluate $|3 \times (-7)| + |13 - 4|$ (1)

✓ (f) Simplify $7 - (x + 2) + (3 - 2x)$ (2)

(g) Simplify $\frac{a^3 - 27}{a^2 - 9}$ (2)

Question 2 (14 marks)**Start a new booklet**

(a) Express the recurring decimal $1.\dot{2}$ as a fraction in simplest terms. (2)

(b) Evaluate $(7.16 \times 10^3) \div (4.94 \times 10^8)$ and give your answer in scientific notation correct to 3 significant figures. (2)

(c) If $a = -2$, $b = 6$ and $c = -1$, evaluate $\frac{12a^2}{b+2c}$ (1)

(d) Rationalise the denominator: $\frac{8}{\sqrt{7}+1}$ (2)

(e) Simplify $\frac{x+7}{3} + \frac{2x-1}{2}$ (2)



(f) Solve $\frac{6}{x} + 2 = \frac{7}{2x} + 5$ (2)

(g) Solve and graph your solution on the number line: $\left| \frac{3x-1}{2} \right| > 1$ (3)

Question 3 (14 marks)**Start a new booklet**

(a) Show that $f(x) = 2x^2 + 5$ is an even function.

(2)

(b) Simplify $\left(\frac{8}{27}\right)^{\frac{1}{3}} \times \left(\frac{4}{9}\right)^{-\frac{1}{2}}$

(1)

(c) If $\sqrt{28} + \sqrt{7} = \sqrt{A}$ find A .

(1)

(d) Express in simplest form without negative or fractional indices: $(8x^6)^{\frac{1}{3}} x^{-3}$

(2)

(e) If $f(x) = 2^x + 2^{-x}$, find $f(1)$ and $f(-2)$.

(2)

(f) What is the domain for:

(4)

(i) $f(x) = \frac{1}{x+1}$

(ii) $f(x) = \sqrt{25 - x^2}$

(iii) $f(x) = \sqrt{6 - x}$

(iv) $f(x) = 2^x$

$y = \sqrt{6} - \sqrt{x}$
 ~~$y = \sqrt{6} - x^{\frac{1}{2}}$~~

(g) Given the function:

$$f(x) = \begin{cases} 2 - x^2 & \text{for } x \leq 0 \\ x^2 - 2 & \text{for } x > 0 \end{cases}$$

Find:

- (i) $f(0)$
(ii) $f(a^2)$

(2)

Question 4 (14 marks)**Start a new booklet**

Using a pencil and quarter page diagrams, sketch the following graphs showing all significant features. State the domain and range in each case.

(i) $f(x) = 3x + 2$ (2)

(ii) $f(x) = x^2 - 2$ (3)

(iii) $f(x) = \frac{2}{x}$ (3)

(iv) $f(x) = |x - 1|$ (3)

(v) $f(x) = \sqrt{9 - x^2}$ (3)

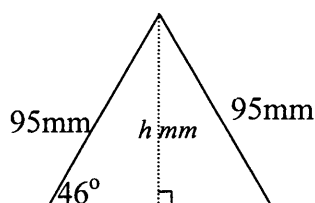
Question 5 (14 marks) **Start a new booklet**

(a) Solve $-2 < 4 - 2x \leq 6$ (2)

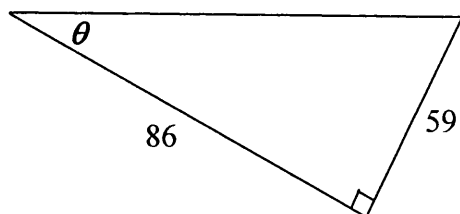
(b) If $\cos A = \frac{2}{7}$ and A is acute, find the exact value of $\sin A$ (2)

(c) Find the value of x if $\cos x^\circ = \sin 20^\circ$ (1)

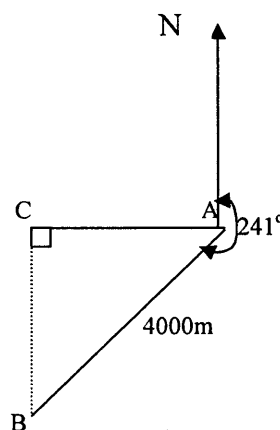
(d) Find the value of h to the nearest whole number. (2)



(e) Find θ in degrees and minutes, to the nearest minute. (2)

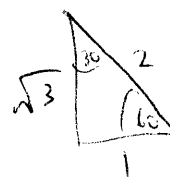


(f) The bearing of B from A is 241° and B is 4000 metres from A . Find AC , which is the distance that B is west of A . (2)



(g) Give the exact value of $\sin 30^\circ \cos 60^\circ$. (2)

(h) Solve $\sin x = \frac{\sqrt{3}}{2}$ for $0^\circ \leq x \leq 90^\circ$ (1)



Question 6 (14 marks)**Start a new booklet**

- (a) Two walkers set off at the same time from a cross road and walk along flat, straight roads inclined to each other at 68° . If they both walk at a speed of 6 km/h , find their distance apart 10 minutes later. (3)
- (b) Sketch the region in the xy plane where $y \geq x^2 + 1$ and $y \leq x + 3$ hold simultaneously. (3)
- (c) Solve simultaneously: $y = 3x + 10$ (4)
 $y = x^2 + x - 5$
- (d) A person, 2m tall, is standing on the ground and looking up at the top of a building. The person is 18m from the building and the angle of elevation to the top of the building is 30° . Draw a diagram representing this information and show that the exact height of the building is $6\sqrt{3} + 2$ metres. (4)

Suggested Solutions

Comments

Question 1

$$\begin{aligned} a) \quad 4\sqrt{45} - 3\sqrt{54} - 6\sqrt{5} &= 12\sqrt{5} - 9\sqrt{6} - 6\sqrt{5} \quad (1) \\ &= 6\sqrt{5} - 9\sqrt{6} \quad (1) \end{aligned}$$

$$\begin{aligned} b) \quad 6(y-1) &= 3(y+8) & 6y-6 &= 3y+24 \quad (1) \\ & & 3y &= 30 \\ & & y &= 10 \quad (1) \end{aligned}$$

$$\begin{aligned} c) \quad 5x-2y &= -1 \quad (1) & (2) \times 2: 16x-2y &= 10 \\ 8x-y &= 5 \quad (2) & (1): 5x-2y &= -1 \quad (1) \\ & & (2) - (1) & \quad 11x = 11 \\ & & & \quad x = 1 \quad (3) \quad (1) \end{aligned}$$

$$\begin{aligned} \text{Substitute (3) into (1)} \quad 5 \times 1 - 2y &= -1 \\ -2y &= -6 \\ y &= 3 \quad (1) \quad x=1, y=3 \end{aligned}$$

$$\begin{aligned} d) \quad \sqrt[3]{\frac{84 \times 69}{35+37}} &= 4.31783 \quad (1) \\ &= 4.32 \quad (1) \quad (3 \text{ sig figs}) \end{aligned}$$

$$\begin{aligned} e) \quad |3 \times (-7)| + |13-4| &= |-21| + |9| \\ &= 21 + 9 \\ &= 30 \quad (1) \end{aligned}$$

$$\begin{aligned} f) \quad 7 - (x+2) + (3-2x) &= 7-x-2+3-2x \quad (1) \\ &= 8-3x \quad (1) \end{aligned}$$

$$\begin{aligned} g) \quad \frac{a^3-27}{a^2-9} &= \frac{(a-3)(a^2+3a+9)}{(a-3)(a+3)} \quad (1) \\ &= \frac{a^2+3a+9}{a+3} \quad (1) \end{aligned}$$

Suggested Solutions

Comments

Question 2

$$a) \text{ let } x = 1.2222 \dots \quad (1)$$

$$10x = 12.2222 \dots \quad (2)$$

$$(2) - (1): 9x = 11$$

$$x = \frac{11}{9} \text{ or } 1\frac{2}{9}$$

$$b) (7.16 \times 10^3) \div (4.94 \times 10^8) = 0.000014493 \dots$$

$$= 1.44939 \dots \times 10^{-5}$$

$$c) \frac{12a^2}{b+2c} = \frac{12 \times (-2)^2}{6+2 \times -1}$$

$$= \frac{48}{4}$$

$$= 12$$

$$d) \frac{8}{(\sqrt{7}+1)} \times \frac{(\sqrt{7}-1)}{(\sqrt{7}-1)} = \frac{8(\sqrt{7}-1)}{7-1}$$

$$= \frac{4(\sqrt{7}-1)}{3}$$

$$e) \frac{(x+7)}{3} \times \frac{2}{2} + \frac{(2x-1)}{2} \times \frac{3}{3} = \frac{2x+14+6x-3}{6}$$

$$= \frac{8x+11}{6}$$

$$f) \frac{6 \times (2x)}{x} + 2 \times (2x) = \frac{7 \times (2x)}{2x} + 5 \times (2x)$$

$$12 + 4x = 7 + 10x$$

$$5 = 6x$$

$$x = \frac{5}{6}$$

$$g) \left| \frac{3x-1}{2} \right| > 1$$

$$\frac{3x-1}{2} > 1 \text{ or } \frac{3x-1}{2} < -1$$

$$3x-1 > 2$$

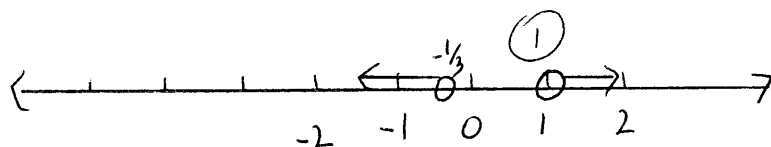
$$3x > 3$$

$$x > 1 \quad (1)$$

$$3x-1 < -2$$

$$3x < -1$$

$$x < -\frac{1}{3} \quad (2)$$



Suggested Solutions

Comments

Question 3

$$\begin{aligned} \text{a) } f(-x) &= 2 \times (-x)^2 + 5 \quad (1) \\ &= 2x^2 + 5 \\ &= f(x) \end{aligned} \quad \begin{aligned} f(x) &= f(-x) \quad (1) \\ \therefore f(x) &\text{ is even} \end{aligned}$$

$$\begin{aligned} \text{b) } \left(\frac{8}{27}\right)^{\frac{1}{3}} \times \left(\frac{4}{9}\right)^{-\frac{1}{2}} &= \frac{2}{3} \times \frac{3}{2} \quad (1) \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{c) } \sqrt{28} + \sqrt{7} &= 2\sqrt{7} + \sqrt{7} \quad (1) \\ &= 3\sqrt{7} \quad (1) \\ &= \sqrt{63} \quad A=63 \end{aligned}$$

$$\begin{aligned} \text{d) } (8x^6)^{\frac{1}{3}} x^{-3} &= 2x^2 \times x^{-3} \quad (1) \\ &= \frac{2}{x} \quad (1) \end{aligned}$$

$$\begin{aligned} \text{e) } f(1) &= 2^1 + 2^{-1} \quad f(-2) = 2^{-2} + 2^2 \\ &= 2 + \frac{1}{2} \quad = \frac{1}{4} + 4 \\ &= 2\frac{1}{2} \quad (1) \quad = 4\frac{1}{4} \quad (1) \end{aligned}$$

$$\begin{aligned} \text{f) i) } x &\neq -1, \text{ otherwise } x \text{ is any real number} \quad (1) \\ \text{ii) } -5 &\leq x \leq 5 \quad (1) \\ \text{iii) } x &\leq 6 \quad (1) \\ \text{iv) } x &\text{ can be any real number} \quad (1) \end{aligned}$$

$$\begin{aligned} \text{g) i) } f(0) &= 2 - 0 \\ &= 2 \quad (1) \end{aligned}$$

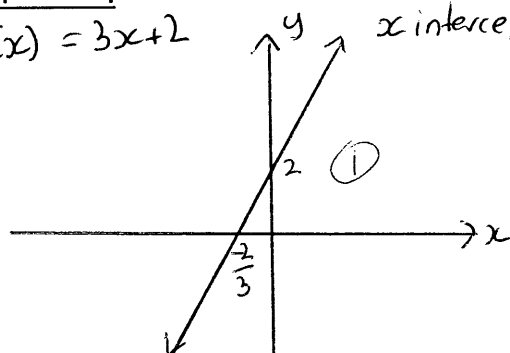
$$\begin{aligned} \text{ii) } f(a^2) \quad (a^2 > 0 \text{ for all } a) \\ &= (a^2)^2 - 2 \\ &= a^4 - 2 \quad (1) \end{aligned}$$

Suggested Solutions

Comments

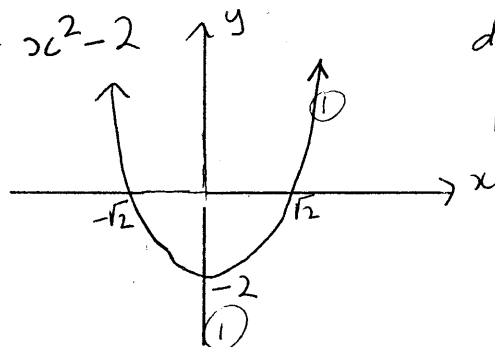
Question 4

i) $f(x) = 3x + 2$

x intercept: $-\frac{2}{3}$, y intercept: 2domain: x is any real numberrange: y is any real number

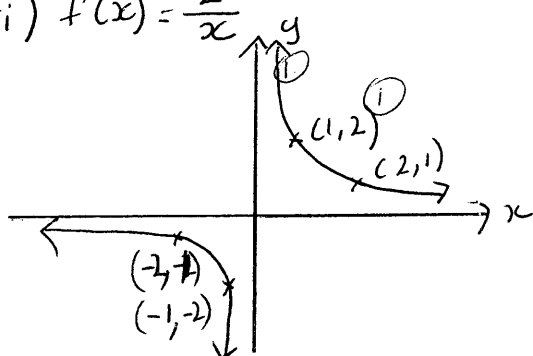
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ii) $f(x) = x^2 - 2$

domain: x is any real numberrange: $y \geq -2$

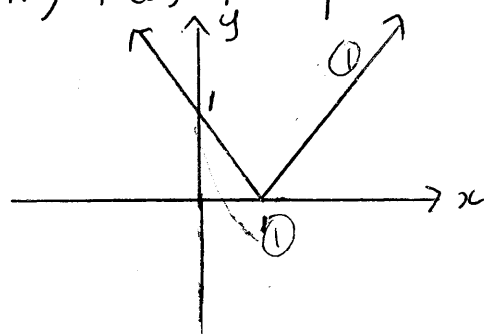
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iii) $f(x) = \frac{2}{x}$

domain: $x \neq 0$ range: $y \neq 0$

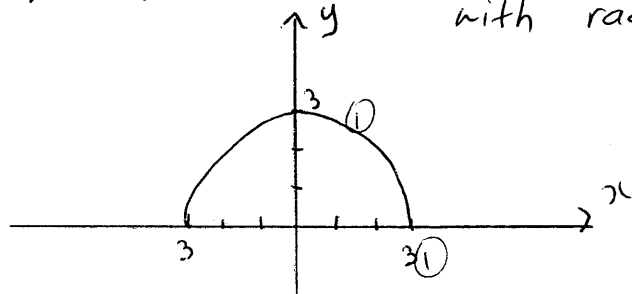
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iv) $f(x) = |x - 1|$

domain: x is any real numberrange: $y \geq 0$

3

v) $f(x) = \sqrt{9 - x^2}$

top half of circle centre (0, 0)
with radius = 3.domain: $-3 \leq x \leq 3$ range: $0 \leq y \leq 3$

3

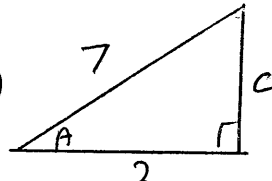
Suggested Solutions

Comments

Question 5

a) $-2 < 4 - 2x \leq 6$ $\begin{matrix} -4 \\ \div -2 \end{matrix}$ $-6 < -2x \leq 2$ ①
 $3 > x \geq -1$ or $-1 \leq x < 3$

2

b)  $c = \sqrt{7^2 - 2^2} = \sqrt{45}$ ①
 $\sin A = \frac{\sqrt{45}}{7}$ or $\frac{3\sqrt{5}}{7}$ ①

2

c) $x = 70^\circ$

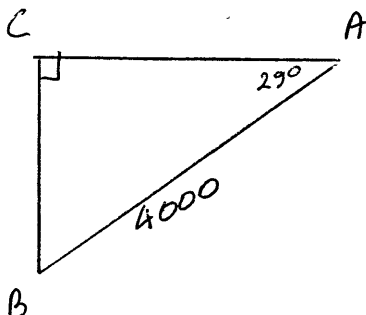
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d) $\sin 46^\circ = \frac{h}{95}$ $h = 95 \sin 46^\circ$ ①
 $= 68.337$
 $= 68 \text{ mm}$ ①

2

e) $\tan \theta = \frac{59}{86}$ ①
 $\theta = 34.45^\circ$
 $= 34^\circ 27'$ ①

2

f)  $\angle A = 360^\circ - 241^\circ - 90^\circ = 29^\circ$ ①
 $\frac{AC}{4000} = \cos 29^\circ$ ① any rounding
 $AC = 3498.478829 \text{ m}$ (6 dp)

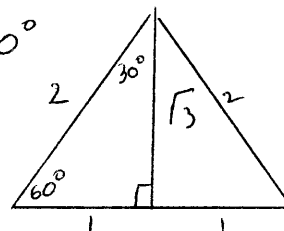
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g) $\sin 30^\circ \cos 60^\circ = \frac{1}{2} \times \frac{1}{2}$ ① ①
 $= \frac{1}{4}$

2

h) $\sin x = \frac{\sqrt{3}}{2}$ for $0^\circ \leq x \leq 90^\circ$

$x = 60^\circ$ ①

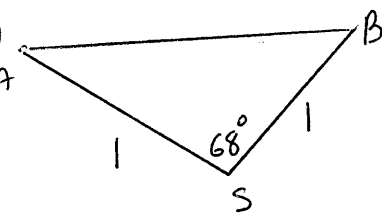


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Suggested Solutions

Comments

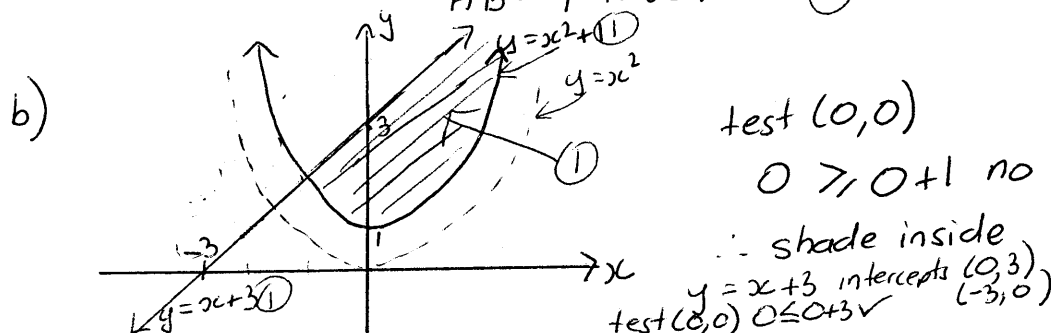
Question 6

a)  after 10 mins travelled $\frac{10}{60} \times 6 = 1$ km ① 3

$$AB^2 = 1^2 + 1^2 - 2 \times 1 \times 1 \cos 68^\circ \quad \text{①}$$

$$AB^2 = 1.250787 \quad \text{①}$$

$$AB = 1.11839 \text{ km} \quad \text{① (unrounding)}$$



c) $y = 3x + 10$ ① $y = x^2 + x - 5$ ② Sub ① into ②: $3x + 10 = x^2 + x - 5$ 4

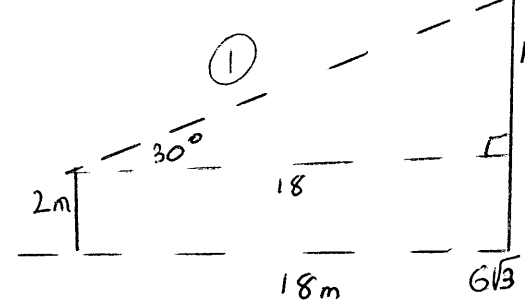
$$0 = x^2 - 2x - 15$$

$$0 = (x - 5)(x + 3)$$

$$x = 5 \text{ or } -3$$

sub ③ into ① $y = 3 \times 5 + 10$ or $y = 3 \times -3 + 10$
 $= 25$ $= 1$

solution: $(5, 25)$ or $(-3, 1)$

d)  $\tan 30^\circ = \frac{h}{18}$ ①
 $\frac{1}{\sqrt{3}} = \frac{h}{18}$
 $h = \frac{18}{\sqrt{3}}$ ① or $\frac{18 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}} = 6\sqrt{3}$
 building height ①
 $= \frac{18}{\sqrt{3}} + 2$
 $= \frac{18\sqrt{3}}{3} + \frac{6}{3}$
 $= \frac{18\sqrt{3} + 6}{3}$ ①
 $= 6\sqrt{3} + 2$ m